Sounak Paul

Curriculum Vitae

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Birthplace: Purulia, India
Date of Birth: 18 Oct 1996

Education

2017– **Master of Science in Mathematics (Thesis based)**, *University of Alberta*.
GPA: 4.0/4.0

2014–2017 **Bachelor of Mathematics (Honours)**, *Indian Statistical Institute Bangalore*.
First Division with Distinction (Class Rank: 2) | Rank Certificate

Secured aggregate of 88.71% overall, and 91.2% in 16 core courses.

Research Experience

May 2018– **Topic:** **Long Range Dependence and Heavy Tails in Data (Masters Thesis)**,
Supervisor: Michael Kouritzin, University of Alberta.

- Used Marcinkiewicz strong laws of large numbers (MSLLN) to improve detetection of long range dependence coefficient σ for certain estimators based on two-sided increasing lag linear models.
- Generalized MSLLN results of well known autocovariance case to multiple products of Heavy-Tailed long range dependent data. I have found that a MSLLN with a faster rate exists for the triple product of the data. However, my calculations also indicate that this will eventually not be true for higher even products (though in some cases with minor extra assumptions, might hold for higher odd products).
- Currently investigating necessary central limit theorems and non-central limit theorems (in terms of stable and Rosenblatt processes) to assert optimality of the rates, and planning to study large deviation principles for these stochastic approximation algorithm, along with applications.

May–Jul 2017 **Topic:** **Extensions of Morphological Operators to Weighted Graphs and applications to Morphological Interpolation**, | Link
Supervisor: BS Daya Sagar, ISI Bangalore.

- Conducted in Systems Science and Informatics Unit (SSIU) of ISI, under the KVPY fellowship.
- Extended notions of dilations, erosions, opening, closing, and other Morphological operators to vertex and edge weighted graphs, and formulated a notion of medians in these graphs via morphological operators.
- Formulated a method of interpolation using successive medians, following ideas from supervisor's previous papers, and simulated on various datasets in Python.

May–Jul 2016 **Topic:** **Special Topics in Analysis and Partial Differential Equations**, | Link
Guide: K Sandeep, TIFR CAM.

- Advanced reading project in PDE and related areas conducted under the TIFR CAM VSRP Fellowship.
- Reviewed basic concepts of Functional Analysis, Operator Theory and Geometric Analysis, then studied Frechet and Gateux derivatives.
- Conducted a comprehensive study of stability theory and distribution theory, with emphasis on distribution solutions of general PDEs from analytic as well as geometric point of view.

May–Jul 2015 **Topic:** **A Proof of SLLN for Bernoulli random variables, and Fekete Lemma**,
Guide: Yogeshwaran D., ISI Bangalore.

Awards and Achievements

2018 **Dr. Josephine M. Mitchell Scholarship**, University of Alberta.

2017 **Pundit RD Sharma Memorial Graduate Award**, University of Alberta, | Link.

2017 **University of Alberta Master's Scholarship**, University of Alberta.
Awarded to 20-odd masters students in the entire Faculty of Graduate Studies and Research. | Link

2017 **TIFR CAM Graduate Scholarship**, *(Declined)*, | Link.

2016 **Visiting Student Research Program Fellowship**, TIFR Bombay & TIFR CAM.
Was one of 17 students to be selected by TIFR Center for Applicable Mathematics, and one of 26 students selected by TIFR Bombay from all over India for a two-month project. | Link

2016 **Summer Research Program Fellowship**, Indian Academy of Sciences, | Link.

2016 **Book Grants for 3 semesters**, ISI Bangalore.

2016 **Joint Admission Test to M.Sc (IIT-JAM)**.
Secured All India Rank 68 for admission to Math MSc Programs at the IITs, and IISc Bangalore.

2014 **West Bengal Joint Entrance Exam:** Secured rank 219 out of 1.7 lakh candidates.

2013 **Regional Mathematics Olympiad (RMO) Awardee:** Top 30 in West Bengal Region.

2013 **Kishore Vaigyanik Protsahan Yojana Scholarship**, IISc Bangalore.
Awarded by IISc Bangalore and Dept. of Science and Technology, India. | Link

2006 **Infomatrix Computer Project Competition at Bucharest, Romania**.
Represented India, and was the youngest participant at this event organised by the Ministry of Education, Research and Youth, and Lumina institutii de invatamant, Bucharest. | Link

Seminars, Workshops and Advanced Schools

Sep 2018 **International Conference on SPDEs**, University of Alberta

Sep 2018 **Graduate Teaching and Learning Program**, University of Alberta.

Aug 2018 **PIMS-AMI Workshop on Applied Harmonic Analysis & Statistical Learning**, UofA

Jun 2017 **Advanced Instructional School on Representation Theory of Finite Groups** | Link.
Chennai Mathematical Institute

May 2017 **A Probability Meeting**, ISI Bangalore, *Attended Unofficially*.

Dec 2016 **School on K-theory and its Applications**, ISI Bangalore, *Attended Unofficially*.

2013 **Indian National Mathematical Olympiad Training Camp**, ISI Kolkata

Talks Presented

Fall 2018 **Long Range Dependence and Heavy Tails in Data**, Graduate Colloquium, UofA

Teaching and Volunteering

Aug 2018 Volunteer at **Canadian Mathematical Society Camp**, UAlberta.

2017- Tutor at **Decima Robinson Support Center**

2017- Taught lab classes and graded assignments in courses like MATH 209 (**Calculus III**), STAT 265 (**Statistics I), STAT 266 (Statistics II), and MATH 309 (Complex Analysis).**

Other Information

Skilled in R, Matlab, Python, C, and LaTeX

Was the **General Secretary** of **ISI Bangalore Student Committee** in 2016-17.